

AIM-Spice: For All Your Analogue Simulation Needs

Developed at the University of California and first released in 1972, Automatic Integrated Circuit Modeling Spice, or AIM-Spice, uses SPICE format for describing analogue circuits. Unlike SPICE, which uses control records, AIM-Spice uses point-and-click type controls. This article focuses on AIM-Spice, which is included in this month's EFY DVD

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SPICE is a well-known, widely documented and standard analogue simulator in the electronics industry. A general-purpose analogue simulator, SPICE can handle complex non-linear circuits and also contains models for most of the circuit elements. Calculation of DC operating points, transient and distortion analyses, locating poles and zeros for different kinds of transfer functions, finding the small signal frequency response and small signal transfer functions, and performing Fourier functions are a few functionalities of the simulator.

Compatible with Microsoft Windows and Linux operating systems, AIM-Spice is a new version of SPICE. The AIM-Spice simulation package consists of two main applications—AIM-Spice itself and a graphic post-processor called AIM-Postprocessor.

The development of AIM-Spice was motivated by the need for a new set of advanced device models for circuit simulation and a more user-friendly interface. For Windows, AIM-Spice is capable of displaying graphically the results of a simulation in progress—a feature that allows the user to terminate a run based on an instant information on intermediate simulation results.

Features

Running on Microsoft Windows and Linux operating systems, AIM-Spice uses the Berkeley SPICE version 3.E1 as the kernel. It displays the simulation

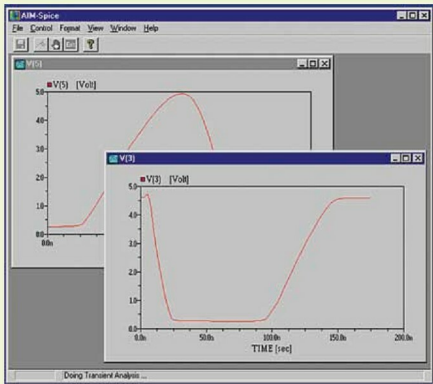


Fig. 1: A typical snapshot of an AIM-Spice simulation in progress

results in progress by plotting graphically circuit variables during a run. It includes advanced new models for a number of semiconductor devices.

AIM-Spice is considered as a simple and effective simulation tool for use with analogue circuits. It uses traditional SPICE-like circuit descriptions but without control records. The simulation control uses simple point-and-click approach. The AIM-Spice help menu is very comprehensive. AIM-Spice toolbars in help section explain all of the icons.

Simulation control

In AIM-Spice for Windows the operator has complete control during a simulation. Before it starts, circuit variables that are to be monitored during the run are selected. AIM-Spice software graphically displays the progress of these variables during the simulation.

Circuit analysis with AIM-Spice

There are different types of analyses in AIM-Spice: